# Diego Garcia

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## Education

**University of Missouri - Columbia** Columbia, MO

*Bachelor of Science in Computer Science, Minor in Mathematics - GPA 3.5* *August 2022 – May 2026*

- **Relevant Coursework:** Cloud Computing, Big Data Analytics, Database Applications & Information Systems, Artificial Intelligence

## Experience

**Software Engineer** June 2026 – Present

*Tata Consultancy Services* *Cincinnati, OH*

- Designed and shipped a full-stack AI chatbot application with a FastAPI backend and React frontend for the Pinnacle onboarding program, integrating an OpenAI model with a RAG pipeline over training documents to give new hires instant answers and reduce the support workload on trainers.
- Completed technical training in Python, FastAPI, and LangChain focused on building LLM-powered applications and backend services, along with enterprise onboarding on Salesforce platform APIs.

**AI/ML Engineer Intern** March 2026 – June 2026

*LEAP Innovations* *Remote*

- Engineered a six-agent autonomous SDLC platform that reads Jira tickets and delivers production-ready GitHub PRs with zero human involvement using Python, FastAPI, and Azure AI.
- Built real-time event-driven pipeline infrastructure including webhook receivers, SSE streaming, and OAuth integrations for GitHub and Jira.
- Improved agent reliability through harness engineering by adding verification gates, context generation, and static analysis fixes across a multi-language codebase.
- Contributed to the full stack across a TypeScript Next.js frontend and Python FastAPI backend, shipping features and bug fixes across four interconnected repositories.

**Undergraduate Research Assistant** May 2025 – August 2025

*University of Missouri - Columbia* *Columbia, MO*

- Designed and deployed real-time data ingestion pipelines across 10+ containerized DERMS microservices (Docker, Kubernetes), enabling scalable ETL of high-frequency DER telemetry.
- Built streaming ETL pipelines with Apache Kafka and ksqlDB, transforming and processing high-frequency sensor data with low-latency SQL-based stream transformations.
- Automated CI/CD workflows for ML and federated learning models, reducing deployment time and improving reproducibility across Kubernetes clusters.

## Projects

**AI Investment Analyst** | *Python, LangChain, LangGraph, Ollama, Chroma, FastAPI, MCP*

- Built a full-stack investment research app with a FastAPI backend, a four-page vanilla JavaScript frontend and a SQLite database, running entirely locally with no external services.
- Implemented real-time NDJSON streaming so users watch an AI agent's tool calls and answers as they happen, and a multi-step trade workflow that pauses for user approval and survives page refreshes.
- Enforced trading rules in deterministic Python using cent-based arithmetic, re-validating every trade before execution and applying it in a single database transaction.
- Wrote 110 pytest tests against an isolated test database, using a scripted fake LLM to cover every workflow path, business rule and API error code in seconds.

**Stock Market Dashboard with ML Predictions** | *FastAPI, PostgreSQL, SQLAlchemy, scikit-learn, Chart.js*

- Built a FastAPI REST API with async endpoints and PostgreSQL integration, caching stock data in a 10-minute window to avoid refetching on every request.
- Designed a SQLAlchemy ORM schema of 3 normalized tables for price history, predictions and model versions, supporting a backfill of 1,000+ days of historical data.
- Developed a responsive JavaScript frontend with Chart.js visualizations, auto-refresh, and live display of ML predictions with confidence scores.

## Technical Skills

**Languages**: Python, SQL, TypeScript, JavaScript, Java, C/C++, C#, R
**Backend**: FastAPI, Node.js, Express, .NET, REST APIs, SQLAlchemy, pytest
**Frontend**: React, Next.js, HTML/CSS, Chart.js, Recharts
**Databases**: PostgreSQL, MySQL, MongoDB, SQLite
**Cloud & DevOps:**: Docker, Kubernetes, CI/CD, Azure, AWS, GCP
**AI**: LangChain, LangGraph, RAG, Azure AI